

IPAD V2.0 COMING SOON?

Going by orders and shipping schedules that Taiwanese component manufacturers are receiving, there could soon be a new version of the segment defining iPad in the offing. These iPads are likely to be even thinner, with ultra-thin, glass-based touch panels
<http://bit.ly/aP2C6v>

HEALTH PACKS NOW A REALITY

How often have we used health packs in games and cribbed about how unrealistic they are. Not any more! Fabrican an instant, sprayable, non-woven fabric will be used in the future to create instant, sprayable, bandages.
<http://bit.ly/bhLOy4>

FARMVILLE: TWISTED EDITION

Say hello to Meet Eater a plant on Facebook that gets watered only when people 'Like' its page. Its first two incarnations died from water overdoses. People come up with weirdest things!
<http://bit.ly/9rgcZy>

**WIRELESS CHARGING NOW A REALITY**

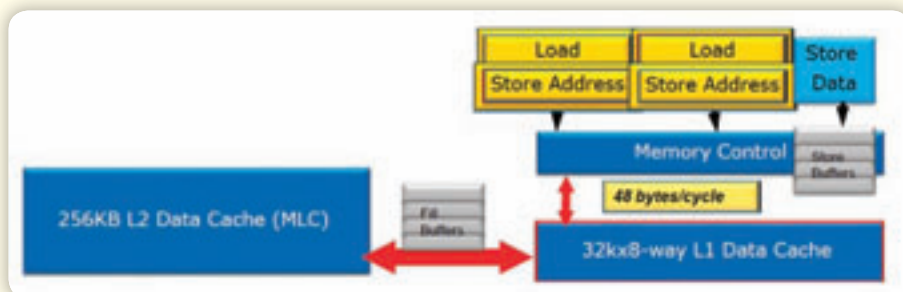
A while ago we'd covered wireless power transmission in its experimental stages – non practical things like charging mats and the like. Fujitsu has taken this to a whole new level where even city-wide wireless networks can be envisioned.
<http://bit.ly/c6W7Qx>

Last level cache

The last level cache (LLC) is shared by the GPU and a special video transcoding engine, and has less latency than we saw in Westmere. It also runs at the core clock speed, which means it's going to be a lot faster than it's been previously. (At least under most circumstances. If the processor cores are downclocked for whatever reason, the LLC will be as well.)

Each core gets its own portion of the LLC (although each core is capable of addressing the entire cache as well), with each portion having its own stop on the ring and a full cache pipeline. This is different from Westmere, in which one cache pipeline was shared by all the cores. According to Intel, the multibanked LLC cache can deliver up to 384 Gbps on a four-core system or 192 Gbps with two cores, with short access latency.

The LLC is logically shared among cores, graphics, and media, with the graphics driver controlling which graphic streams are cached and are coherent. (There are mechanisms in place to prevent thrashing between the core and graphics, and multiple coherency domains for power optimization and anti-snooping purposes.) Because this reduces DRAM accesses, this frees up DRAM bandwidth for the cores and lets the DRAM utilize less power.

**System agent**

The System Agent (formerly known as the uncore) offers a number of features and providing extensive power and thermal management for the PCIe and DDR controllers. The PCIe x16 lane may be split into two x8s, if desired. There's a redesigned dual-channel DDR3 memory controller. And the System Agent also contains a Power Control Unit (PCU), a programmable uController that handles all the power management and reset functions within the chip. The System Agent is fully integrated with the ring, and helps provide the cores, video, and media processing with higher bandwidth and lower latency. It also handles IO-to-cache coherency, and operates on a separate voltage and frequency from the ring, cores, and display.

Improved integrated graphics

One of the Sandy Bridge changes that received the most discussion at IDF was its new graphics hardware. Intel says its goal was to increase the performance of the graphics while maintaining the same socket power, and in order to make that happen the company had to make a "radical departure" from what it was doing before. One Intel rep even said that the change in graphics approach was responsible for the biggest power savings in any part of Sandy Bridge.

